Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Couchman Farms

Physical address of dairy:

3107 Illinois AVE	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/01/1964

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X036-X017-XXXX

B. OPERATORS

Couchman, David Brian

Operator name: Couchman, David Brian

Telephone no.: (209) 527-4956 (Landline) (Cellular)

3136 Iowa AVE	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Couchman, David Brian

Legal owner name: Couchman, David Brian

Telephone no.: (209) 527-4956 (Landline) (Cellular)

3136 Iowa AVE	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Couchman, Mark

Legal owner name:	Couchman, Mark	Telephone no.:	(707) 993-1686	
			Landline	Cellular
351 E Napa ST	Sonoma		CA	95376
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	100	25	0
Number under roof	740	90	190	0	0	0
Maximum number	740	90	190	100	25	0
Average number	740	90	190	100	25	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,928.37 *tons per reporting period*

Total nitrogen from manure: 301,353.28 *lbs per reporting period*

After ammonia losses (30% loss applied): 210,947.30 *lbs per reporting period*

Total phosphorus from manure: 50,392.95 *lbs per reporting period*

Total potassium from manure: 145,120.09 *lbs per reporting period*

Total salt from manure: 369,124.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 32,846,800 *gallons*

Total nitrogen generated: 75,741.05 *lbs*

Total phosphorus generated: 14,377.55 *lbs*

Total potassium generated: 40,398.97 *lbs*

Total salt generated: 577,628.98 *lbs*

	32,846,800	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	32,846,800	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 5	Surface water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/26/2023	Solid commercial fertilizer *UN32*	1.75 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
08/15/2023	Solid commercial fertilizer *5-11-6*	0.69 *ton*	Dry-weight	0.1	5.000000	11.000000	6.000000	0.000000
09/01/2023	Solid commercial fertilizer *2-10-6*	0.44 *ton*	Dry-weight	0.1	2.000000	10.000000	6.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,205.39	239.56	135.46	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	1,205.39	239.56	135.46	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	24	24	2	both	X017-X036-X014-XXXX
2	16	16	1	manure	X017-X036-X014-XXXX
3	26	26	1	none	X017-X036-X014-XXXX
4	70	70	2	process wastewater	X017-X036-X018-XXXX
5	15	15	2	both	X017-X036-X023-XXXX
Big	48	48	2	both	X017-X036-X010-XXXX
Front	23	23	2	both	X017-X036-X010-XXXX
New	26	23	2	both	X017-X036-X010-XXXX
Tack	10	4	2	both	X017-X036-X010-XXXX
Totals for areas that were used for application	232	223	15		
Totals for areas that were not used for application	26	26	1		
Land application area totals	258	249	16		

B. CROPS AND HARVESTS

1

Field name: 1

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 24 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	379.34 *ton*	As-is		74.7	5,500.00	1,000.00	8,000.00		9.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	15.81	173.86	31.61	252.89	738.19

1

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	715.93 *ton*	As-is		67.5	2,300.00	800.00	3,700.00		4.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.83	137.22	47.73	220.75	798.86

2

Field name: 2

01/15/2023: Almond, in shell

Crop: Almond, in shell Acres planted: 16 Plant date: 01/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.50	193.50	32.85	209.25	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

3

Field name: 3

3

01/24/2022: Almond, in shell

Crop: Almond, in shell Acres planted: 26 Plant date: 01/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	0.01 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.50	193.50	32.85	209.25	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

4

Field name: 4

10/30/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	836.00 *ton*	As-is		55.6	6,200.00	1,200.00	11,200.00		6.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	11.94	148.09	28.66	267.52	647.98

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	2,045.47 *ton*	As-is		71.2	3,900.00	800.00	5,000.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.22	227.92	46.75	292.21	1,026.71

5

Field name: 5

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 15 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	237.08 *ton*	As-is		70.7	7,500.00	1,200.00	10,900.00		9.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	15.81	237.08	37.93	344.56	882.66

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	447.46 *ton*	As-is		70.0	2,800.00	800.00	4,000.00		4.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.83	167.05	47.73	238.65	862.70

Big

Field name: Big

10/28/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 48 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	915.91 *ton*	As-is		58.4	8,700.00	1,500.00	7,700.00		9.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	19.08	332.02	57.24	293.85	1,514.55

Big

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	1,131.86 *ton*	As-is		70.2	3,500.00	800.00	4,500.00		4.87

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.58	165.06	37.73	212.22	684.43

Front

Field name: Front

10/29/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 23 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	438.87 *ton*	As-is		56.2	6,000.00	1,200.00	5,900.00		7.14

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	19.08	228.98	45.80	225.16	1,193.47

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	403.68 *ton*	As-is		67.5	4,600.00	900.00	4,000.00		4.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	17.55	161.47	31.59	140.41	528.21

New

Field name: New

10/29/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 23 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	438.87 *ton*	As-is		65.4	5,600.00	900.00	5,200.00		8.42

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	19.08	213.71	34.35	198.45	1,111.80

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	773.53 *ton*	As-is		65.7	3,400.00	1,000.00	4,300.00		4.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.63	228.70	67.26	289.23	1,139.73

Tack

Field name: Tack

10/29/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 4 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	76.32 *ton*	As-is		55.4	7,300.00	1,200.00	9,200.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	19.08	278.57	45.79	351.07	1,242.41

Tack

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	134.52 *ton*	As-is		65.7	3,400.00	1,000.00	4,300.00		4.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.63	228.68	67.26	289.22	1,139.67

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/27/2022: Wheat, silage, soft dough

Field name: 1

Crop: Wheat, silage, soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	88.14	10.46	14.90	0.00	164.00 *ton*
Application event totals		88.14	10.46	14.90	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.85	18.90	0.15	462.65	864,000.00 *gal*
Application event totals		110.85	18.90	0.15	462.65	

1 - 05/22/2023: Corn, silage

Field name: 1

Crop: Corn, silage — Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	185.45	57.62	170.15	3,207.59	130.00 *ton*
Application event totals		185.45	57.62	170.15	3,207.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.65	3,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.65	

1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.14	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.14	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.38	2,827,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.38	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.61	2,535,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.61	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.12	2,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.12	

07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.61	2,535,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.61	

1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.61	2,535,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.61	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	30.10	2,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.10	

2 - 01/15/2023: Almond, in shell

Field name: 2

Crop: Almond, in shell

Plant date: 01/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	87.88	10.43	14.86	0.00	109.00 *ton*
Application event totals		87.88	10.43	14.86	0.00	

08/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-6	Solid commercial fertilizer	1.65	3.63	1.98	0.00	
Application event totals		1.65	3.63	1.98	0.00	

09/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2-10-6	Solid commercial fertilizer	1.10	5.50	3.30	0.00	
Application event totals		1.10	5.50	3.30	0.00	

2 - 01/15/2023: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	8.34	0.00	0.00	0.00	8,885,464.00 *gal*
Application event totals		8.34	0.00	0.00	0.00	

3 - 01/24/2022: Almond, in shell

Field name: 3

Crop: Almond, in shell

Plant date: 01/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5-11-6	Solid commercial fertilizer	1.65	3.63	1.98	0.00	
Application event totals		1.65	3.63	1.98	0.00	

10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	8.34	0.00	0.00	0.00	14,438,879.00 *gal*
Application event totals		8.34	0.00	0.00	0.00	

4 - 10/30/2022: Wheat, silage, soft dough

Field name: 4

Crop: Wheat, silage, soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 10/30/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.35	17.28	0.14	422.99	2,304,000.00 *gal*
River	Surface water	0.82	0.00	0.00	0.00	3,840,000.00 *gal*
Application event totals		102.18	17.28	0.14	422.99	

4 - 05/25/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.81	0.00	0.00	0.00	8,415,000.00 *gal*
Application event totals		1.81	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.01	12.96	0.10	317.24	1,728,000.00 *gal*
River	Surface water	1.55	0.00	0.00	0.00	7,200,000.00 *gal*
Application event totals		77.56	12.96	0.10	317.24	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.58	5.99	35.42	344.73	1,134,000.00 *gal*
River	Surface water	1.32	0.00	0.00	0.00	6,142,500.00 *gal*
Application event totals		28.90	5.99	35.42	344.73	

4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.27	7.22	42.73	415.87	1,368,000.00 *gal*
River	Surface water	1.59	0.00	0.00	0.00	7,410,000.00 *gal*
Application event totals		34.86	7.22	42.73	415.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.96	6.94	41.04	399.45	1,314,000.00 *gal*
River	Surface water	1.53	0.00	0.00	0.00	7,117,500.00 *gal*
Application event totals		33.48	6.94	41.04	399.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.96	6.94	41.04	399.45	1,314,000.00 *gal*
River	Surface water	1.53	0.00	0.00	0.00	7,117,500.00 *gal*
Application event totals		33.48	6.94	41.04	399.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.39	7.03	41.60	404.92	1,332,000.00 *gal*
River	Surface water	1.55	0.00	0.00	0.00	7,215,000.00 *gal*
Application event totals		33.94	7.03	41.60	404.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.83	7.13	42.17	410.40	1,350,000.00 *gal*
River	Surface water	1.57	0.00	0.00	0.00	7,312,500.00 *gal*
Application event totals		34.40	7.13	42.17	410.40	

5 - 10/27/2022: Wheat, silage, soft dough

5 - 10/27/2022: Wheat, silage, soft dough

Field name: 5

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	88.57	10.51	14.97	0.00	103.00 *ton*
Application event totals		88.57	10.51	14.97	0.00	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	132.37	22.56	0.18	552.43	644,800.00 *gal*
River	Surface water	0.84	0.00	0.00	0.00	840,000.00 *gal*
Application event totals		133.21	22.56	0.18	552.43	

5 - 05/22/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	182.60	56.73	167.53	3,158.24	80.00 *ton*
Application event totals		182.60	56.73	167.53	3,158.24	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	44.46	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.46	

5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	46.31	2,250,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.31	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.11	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.11	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.11	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.11	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.52	1,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.52	

07/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	28.80	0.00	0.00	0.00	
Application event totals		28.80	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	34.58	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.58	

5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.42	1,575,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.42	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	32.11	1,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.11	

Big - 10/28/2022: Wheat, silage, soft dough

Field name: Big

Crop: Wheat, silage, soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	88.95	10.56	15.04	0.00	331.00 *ton*
Application event totals		88.95	10.56	15.04	0.00	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.72	23.82	0.19	583.13	2,178,000.00 *gal*
River	Surface water	0.68	0.00	0.00	0.00	2,178,000.00 *gal*
Application event totals		140.41	23.82	0.19	583.13	

Big - 10/28/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.01	19.09	0.15	467.47	1,746,000.00 *gal*
River	Surface water	0.55	0.00	0.00	0.00	1,746,000.00 *gal*
Application event totals		112.56	19.09	0.15	467.47	

05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.52	0.00	0.00	0.00	1,656,000.00 *gal*
Application event totals		0.52	0.00	0.00	0.00	

Big - 06/25/2023: Corn, silage

Field name: Big

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.52	13.38	0.11	327.71	1,224,000.00 *gal*
River	Surface water	1.28	0.00	0.00	0.00	4,080,000.00 *gal*
Application event totals		79.80	13.38	0.11	327.71	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.13	9.15	54.11	526.67	1,188,000.00 *gal*
River	Surface water	1.24	0.00	0.00	0.00	3,960,000.00 *gal*
Application event totals		43.37	9.15	54.11	526.67	

Big - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.94	8.46	50.01	486.77	1,098,000.00 *gal*
River	Surface water	1.29	0.00	0.00	0.00	4,117,500.00 *gal*
Application event totals		40.23	8.46	50.01	486.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.39	7.90	46.73	454.85	1,026,000.00 *gal*
River	Surface water	1.29	0.00	0.00	0.00	4,132,500.00 *gal*
Application event totals		37.68	7.90	46.73	454.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.11	7.62	45.09	438.89	990,000.00 *gal*
River	Surface water	1.23	0.00	0.00	0.00	3,918,750.00 *gal*
Application event totals		36.34	7.62	45.09	438.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.61	0.00	0.00	0.00	5,130,000.00 *gal*
Application event totals		1.61	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.61	0.00	0.00	0.00	5,130,000.00 *gal*
Application event totals		1.61	0.00	0.00	0.00	

Big - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.56	0.00	0.00	0.00	4,987,500.00 *gal*
Application event totals		1.56	0.00	0.00	0.00	

Front - 10/29/2022: Wheat, silage, soft dough

Field name: Front

Crop: Wheat, silage, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	88.61	10.52	14.98	0.00	158.00 *ton*
Application event totals		88.61	10.52	14.98	0.00	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	212.07	36.15	0.29	885.06	1,584,000.00 *gal*
River	Surface water	1.03	0.00	0.00	0.00	1,584,000.00 *gal*
Application event totals		213.10	36.15	0.29	885.06	

04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.15	0.00	0.00	0.00	1,764,000.00 *gal*
Application event totals		1.15	0.00	0.00	0.00	

05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.03	0.00	0.00	0.00	1,584,000.00 *gal*
Application event totals		1.03	0.00	0.00	0.00	

Front - 06/24/2023: Corn, silage

Field name: Front

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.76	14.79	0.12	362.07	648,000.00 *gal*
River	Surface water	1.41	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		88.17	14.79	0.12	362.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.94	15.62	92.40	899.30	972,000.00 *gal*
River	Surface water	1.32	0.00	0.00	0.00	2,025,000.00 *gal*
Application event totals		73.27	15.62	92.40	899.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.62	13.60	80.42	782.72	846,000.00 *gal*
River	Surface water	1.15	0.00	0.00	0.00	1,762,500.00 *gal*
Application event totals		63.77	13.60	80.42	782.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.80	0.00	0.00	0.00	2,760,000.00 *gal*
Application event totals		1.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.68	0.00	0.00	0.00	2,580,000.00 *gal*
Application event totals		1.68	0.00	0.00	0.00	

Front - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.65	0.00	0.00	0.00	2,520,000.00 *gal*
Application event totals		1.65	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.69	0.00	0.00	0.00	2,587,500.00 *gal*
Application event totals		1.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	0.00	2,418,750.00 *gal*
Application event totals		1.58	0.00	0.00	0.00	

New - 10/29/2022: Wheat, silage, soft dough

Field name: New

Crop: Wheat, silage, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	88.61	10.52	14.98	0.00	158.00 *ton*
Application event totals		88.61	10.52	14.98	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	173.51	29.58	0.24	724.14	1,296,000.00 *gal*
River	Surface water	1.41	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		174.92	29.58	0.24	724.14	

New - 10/29/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.43	0.00	0.00	0.00	2,190,000.00 *gal*
Application event totals		1.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.21	0.00	0.00	0.00	1,860,000.00 *gal*
Application event totals		1.21	0.00	0.00	0.00	

New - 06/24/2023: Corn, silage

Field name: New

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.88	0.00	0.00	0.00	2,880,000.00 *gal*
Application event totals		1.88	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.62	12.73	75.29	732.76	792,000.00 *gal*
River	Surface water	1.29	0.00	0.00	0.00	1,980,000.00 *gal*
Application event totals		59.91	12.73	75.29	732.76	

New - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.63	10.99	65.02	632.84	684,000.00 *gal*
River	Surface water	1.12	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		51.74	10.99	65.02	632.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.95	14.76	87.27	849.34	918,000.00 *gal*
River	Surface water	1.50	0.00	0.00	0.00	2,295,000.00 *gal*
Application event totals		69.45	14.76	87.27	849.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.63	10.99	65.02	632.84	684,000.00 *gal*
River	Surface water	1.12	0.00	0.00	0.00	1,710,000.00 *gal*
Application event totals		51.74	10.99	65.02	632.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.62	12.73	75.29	732.76	792,000.00 *gal*
River	Surface water	1.29	0.00	0.00	0.00	1,980,000.00 *gal*
Application event totals		59.91	12.73	75.29	732.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.44	0.00	0.00	0.00	2,205,000.00 *gal*
Application event totals		1.44	0.00	0.00	0.00	

New - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.35	0.00	0.00	0.00	2,070,000.00 *gal*
Application event totals		1.35	0.00	0.00	0.00	

Tack - 10/29/2022: Wheat, silage, soft dough

Field name: Tack

Crop: Wheat, silage, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	87.07	10.34	14.72	0.00	27.00 *ton*
Application event totals		87.07	10.34	14.72	0.00	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.57	23.62	0.19	578.31	180,000.00 *gal*
River	Surface water	1.58	0.00	0.00	0.00	420,000.00 *gal*
Application event totals		140.15	23.62	0.19	578.31	

04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		2.03	0.00	0.00	0.00	

05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		2.03	0.00	0.00	0.00	

Tack - 06/24/2023: Corn, silage

Field name: Tack

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.80	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		1.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	2.03	0.00	0.00	0.00	540,000.00 *gal*
Application event totals		2.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.93	19.96	118.07	1,149.11	216,000.00 *gal*
River	Surface water	1.35	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		93.28	19.96	118.07	1,149.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.93	19.96	118.07	1,149.11	216,000.00 *gal*
River	Surface water	1.35	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		93.28	19.96	118.07	1,149.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.69	0.00	0.00	0.00	450,000.00 *gal*
Application event totals		1.69	0.00	0.00	0.00	

Tack - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.80	0.00	0.00	0.00	480,000.00 *gal*
Application event totals		1.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.93	19.96	118.07	1,149.11	216,000.00 *gal*
River	Surface water	1.35	0.00	0.00	0.00	360,000.00 *gal*
Application event totals		93.28	19.96	118.07	1,149.11	

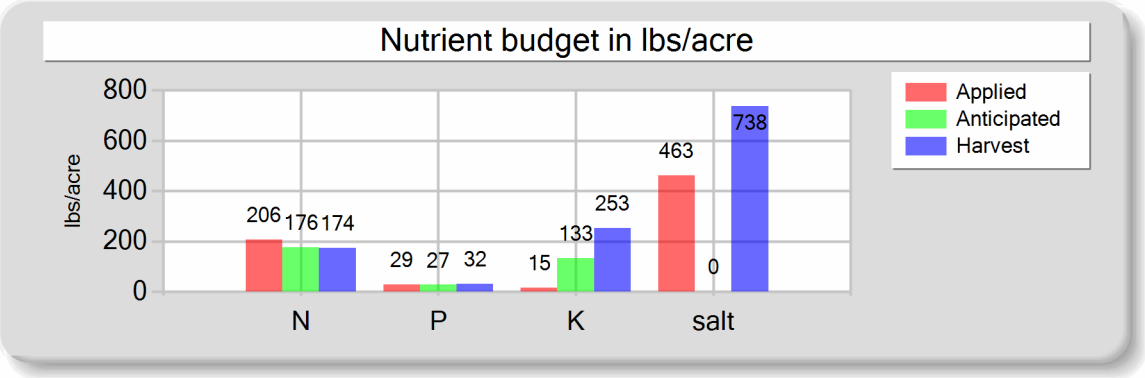
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.46	0.00	0.00	0.00	390,000.00 *gal*
Application event totals		1.46	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/27/2022: Wheat, silage, soft dough

Field name: 1 Crop: Wheat, silage, soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.14	10.46	14.90	0.00
Process wastewater	110.85	18.90	0.15	462.65
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.00	29.36	15.05	462.65
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	173.86	31.61	252.89	738.19
Nutrient balance	32.14	-2.25	-237.84	-275.55
Applied to removed ratio	1.18	0.93	0.06	0.63

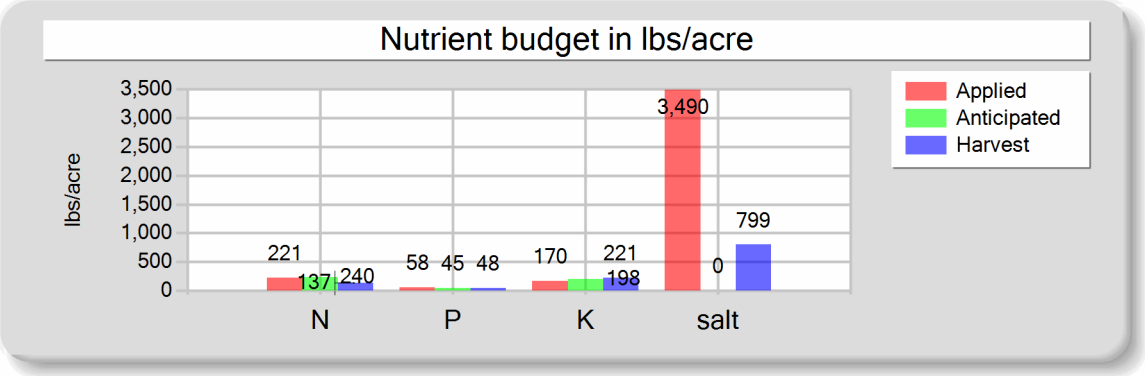
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
864,000.00 *gallons*
31.82 *acre-inches*
1.33 *inches/acre*

Total harvests for the crop
1 *harvests*

1 - 05/22/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	28.80	0.00	0.00	0.00
Dry manure	185.45	57.62	170.15	3,207.59
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	282.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.25	57.62	170.15	3,489.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	137.22	47.73	220.75	798.86
Nutrient balance	84.03	9.89	-50.60	2,690.96
Applied to removed ratio	1.61	1.21	0.77	4.37

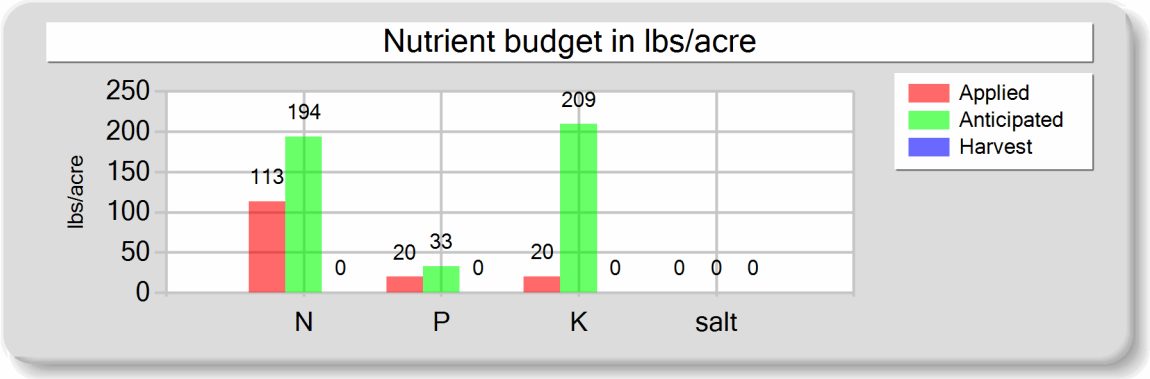
Fresh water applied	
21,937,500.00	*gallons*
807.88	*acre-inches*
33.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 01/15/2023: Almond, in shell

Field name: 2 Crop: Almond, in shell Plant date: 01/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	2.75	9.13	5.28	0.00
Dry manure	87.88	10.43	14.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.34	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	112.97	19.56	20.14	0.00
Anticipated crop nutrient removal	193.50	32.85	209.25	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	112.97	19.56	20.14	0.00
Applied to removed ratio	0.00	0.00	0.00	0.00

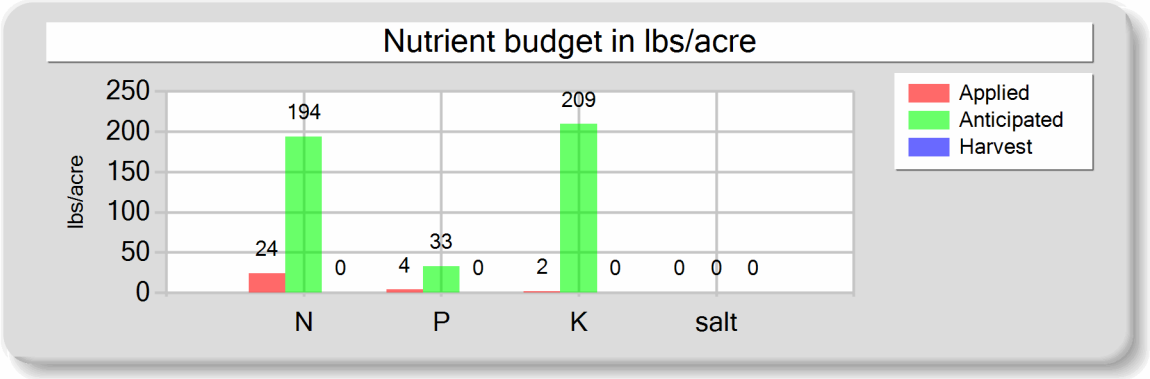
Fresh water applied	
8,885,464.00	*gallons*
327.22	*acre-inches*
20.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 01/24/2022: Almond, in shell

Field name: 3 Crop: Almond, in shell Plant date: 01/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1.65	3.63	1.98	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.34	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	23.99	3.63	1.98	0.00
Anticipated crop nutrient removal	193.50	32.85	209.25	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	23.99	3.63	1.98	0.00
Applied to removed ratio	0.00	0.00	0.00	0.00

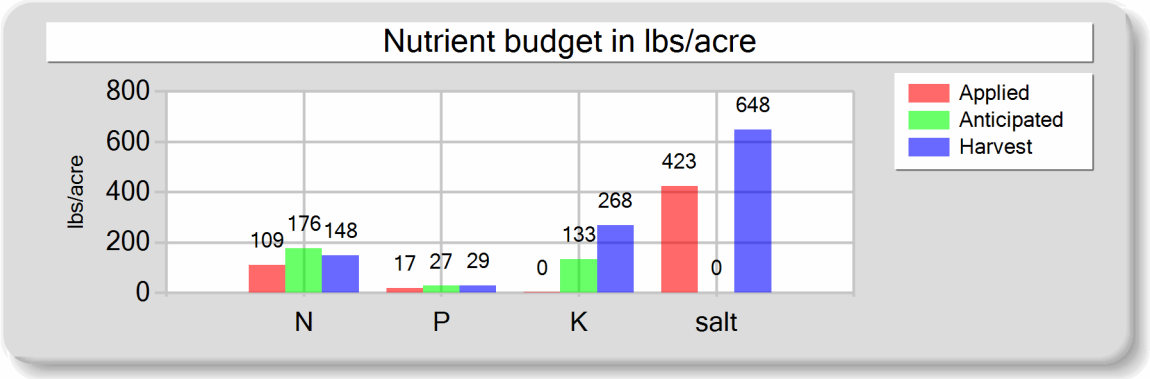
Fresh water applied	
14,438,879.00	*gallons*
531.73	*acre-inches*
20.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/30/2022: Wheat, silage, soft dough

Field name: 4 Crop: Wheat, silage, soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.35	17.28	0.14	422.99
Fresh water	0.82	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	109.18	17.28	0.14	422.99
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	148.09	28.66	267.52	647.98
Nutrient balance	-38.91	-11.39	-267.38	-224.99
Applied to removed ratio	0.74	0.60	0.00	0.65

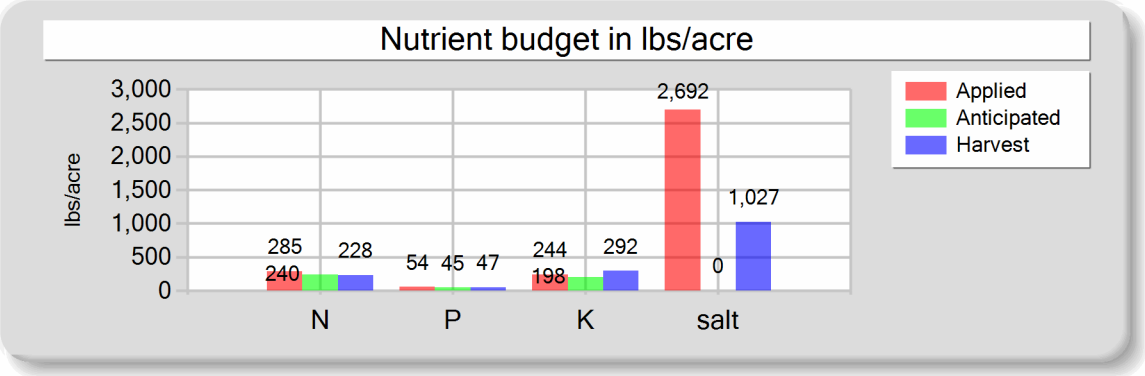
Fresh water applied	
3,840,000.00	*gallons*
141.41	*acre-inches*
2.02	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/25/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	266.00	54.21	244.10	2,692.06
Fresh water	12.43	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.43	54.21	244.10	2,692.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	227.92	46.75	292.21	1,026.71
Nutrient balance	57.51	7.46	-48.11	1,665.35
Applied to removed ratio	1.25	1.16	0.84	2.62

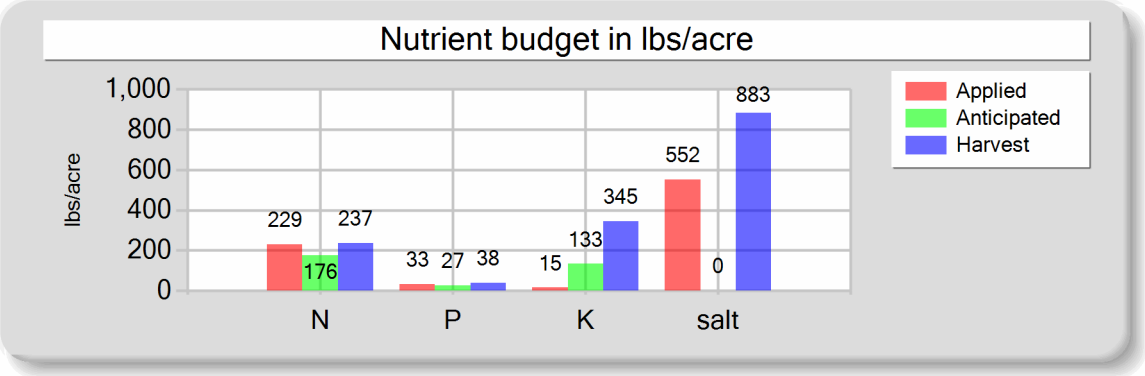
Fresh water applied	
57,930,000.00	*gallons*
2,133.36	*acre-inches*
30.48	*inches/acre*

Process wastewater applied	
9,540,000.00	*gallons*
351.33	*acre-inches*
5.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/27/2022: Wheat, silage, soft dough

Field name: 5 Crop: Wheat, silage, soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.57	10.51	14.97	0.00
Process wastewater	132.37	22.56	0.18	552.43
Fresh water	0.84	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.78	33.08	15.15	552.43
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	237.08	37.93	344.56	882.66
Nutrient balance	-8.30	-4.85	-329.40	-330.23
Applied to removed ratio	0.97	0.87	0.04	0.63

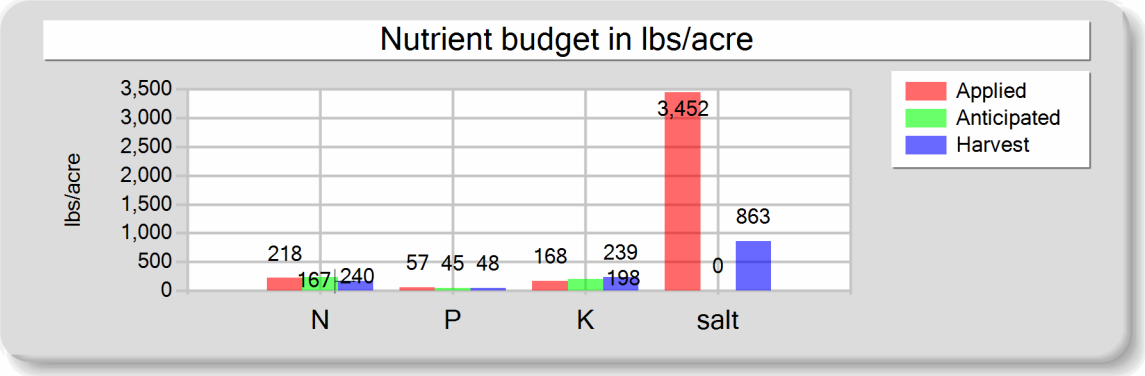
Fresh water applied	
840,000.00	*gallons*
30.93	*acre-inches*
2.06	*inches/acre*

Process wastewater applied	
644,800.00	*gallons*
23.75	*acre-inches*
1.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/22/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	28.80	0.00	0.00	0.00
Dry manure	182.60	56.73	167.53	3,158.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	293.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.40	56.73	167.53	3,451.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	167.05	47.73	238.65	862.70
Nutrient balance	51.35	9.00	-71.12	2,589.18
Applied to removed ratio	1.31	1.19	0.70	4.00

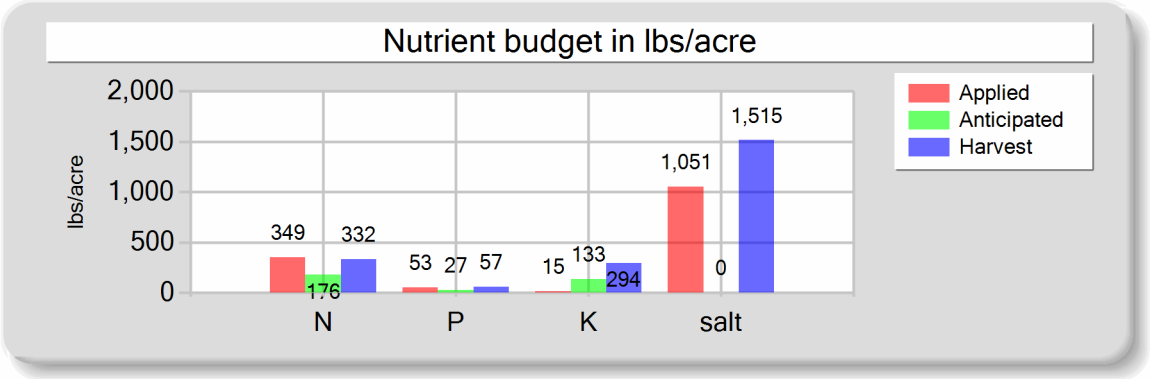
Fresh water applied	
14,265,000.00	*gallons*
525.33	*acre-inches*
35.02	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big - 10/28/2022: Wheat, silage, soft dough

Field name: Big | Crop: Wheat, silage, soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.95	10.56	15.04	0.00
Process wastewater	251.73	42.91	0.34	1,050.59
Fresh water	1.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	349.43	53.47	15.38	1,050.59
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	332.02	57.24	293.85	1,514.55
Nutrient balance	17.41	-3.77	-278.47	-463.96
Applied to removed ratio	1.05	0.93	0.05	0.69

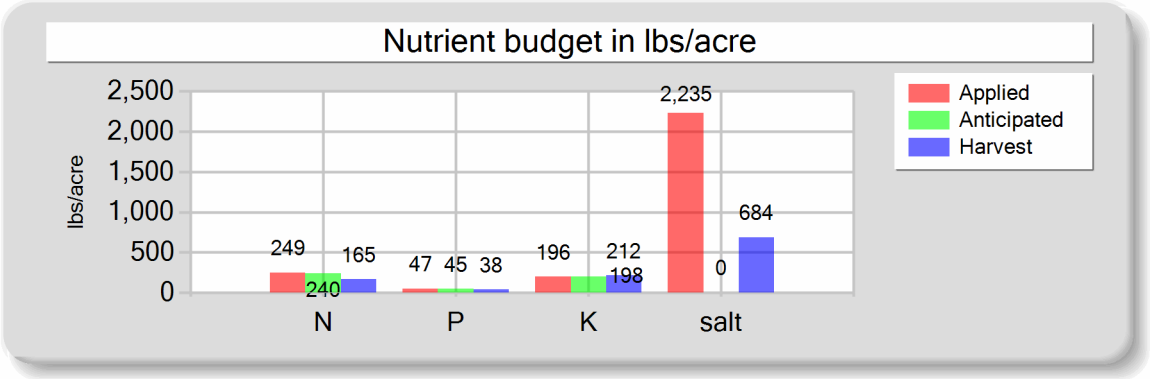
Fresh water applied	
5,580,000.00	*gallons*
205.49	*acre-inches*
4.28	*inches/acre*

Process wastewater applied	
3,924,000.00	*gallons*
144.51	*acre-inches*
3.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big - 06/25/2023: Corn, silage

Field name: Big Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	231.10	46.52	196.06	2,234.91
Fresh water	11.10	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.19	46.52	196.06	2,234.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	165.06	37.73	212.22	684.43
Nutrient balance	84.13	8.79	-16.16	1,550.48
Applied to removed ratio	1.51	1.23	0.92	3.27

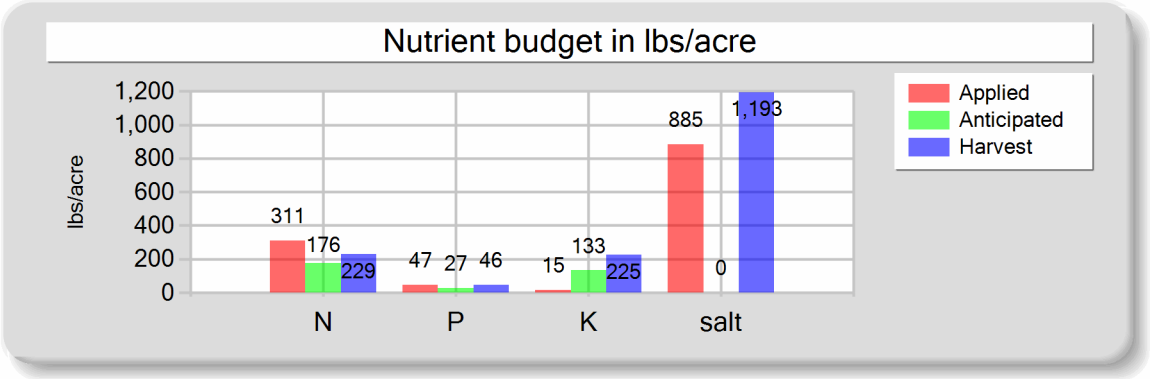
Fresh water applied	
35,456,250.00	*gallons*
1,305.73	*acre-inches*
27.20	*inches/acre*

Process wastewater applied	
5,526,000.00	*gallons*
203.50	*acre-inches*
4.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Front - 10/29/2022: Wheat, silage, soft dough

Field name: Front | Crop: Wheat, silage, soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.61	10.52	14.98	0.00
Process wastewater	212.07	36.15	0.29	885.06
Fresh water	3.22	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.90	46.67	15.27	885.06
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	228.98	45.80	225.16	1,193.47
Nutrient balance	81.93	0.87	-209.89	-308.40
Applied to removed ratio	1.36	1.02	0.07	0.74

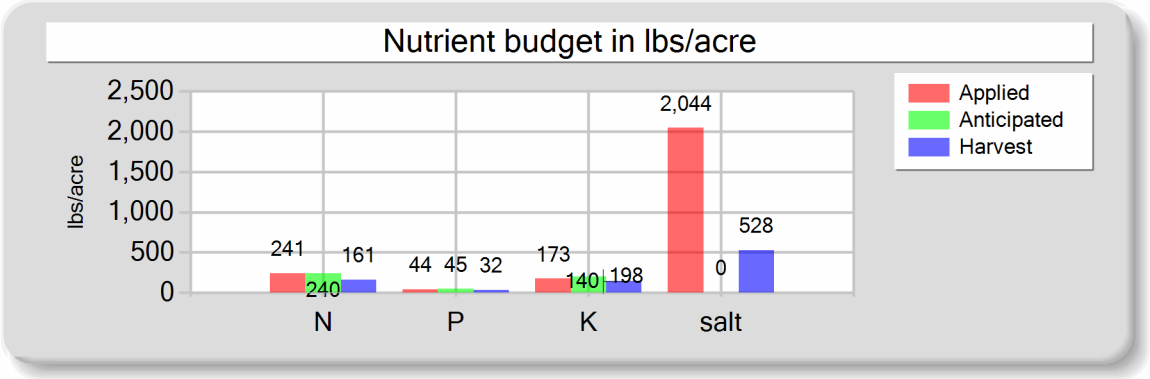
Fresh water applied	
4,932,000.00	*gallons*
181.63	*acre-inches*
7.90	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
2.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Front - 06/24/2023: Corn, silage

Field name: Front | Crop: Corn, silage | Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.32	44.01	172.94	2,044.10
Fresh water	12.29	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.61	44.01	172.94	2,044.10
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	161.47	31.59	140.41	528.21
Nutrient balance	79.13	12.42	32.53	1,515.89
Applied to removed ratio	1.49	1.39	1.23	3.87

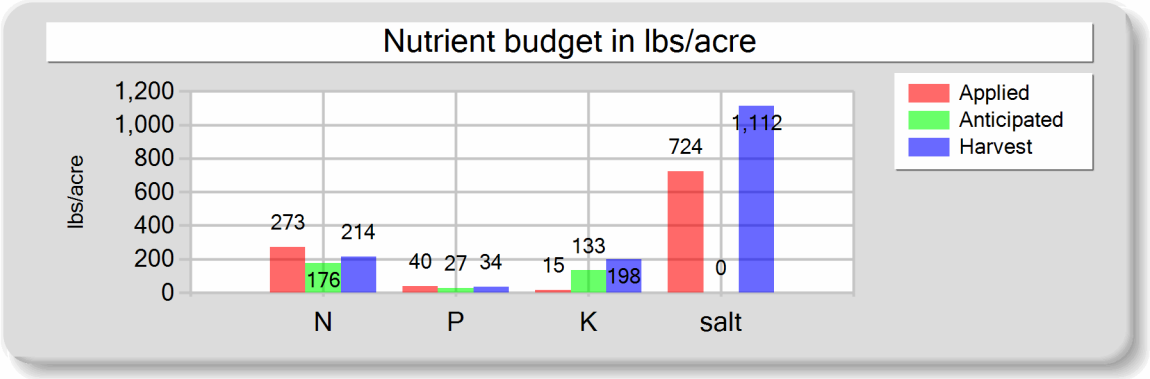
Fresh water applied	
18,813,750.00	*gallons*
692.85	*acre-inches*
30.12	*inches/acre*

Process wastewater applied	
2,466,000.00	*gallons*
90.81	*acre-inches*
3.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

New - 10/29/2022: Wheat, silage, soft dough

Field name: New | Crop: Wheat, silage, soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.61	10.52	14.98	0.00
Process wastewater	173.51	29.58	0.24	724.14
Fresh water	4.06	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.18	40.10	15.22	724.14
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	213.71	34.35	198.45	1,111.80
Nutrient balance	59.47	5.75	-183.23	-387.66
Applied to removed ratio	1.28	1.17	0.08	0.65

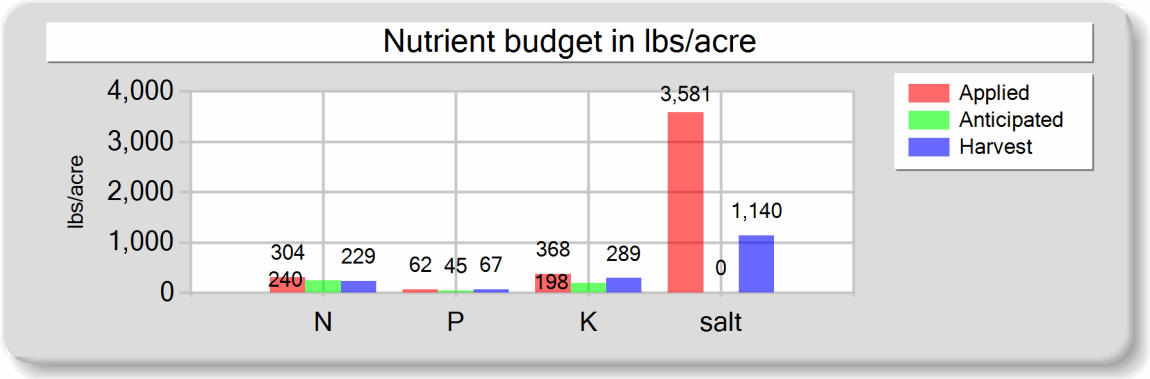
Fresh water applied	
6,210,000.00	*gallons*
228.69	*acre-inches*
9.94	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

New - 06/24/2023: Corn, silage

Field name: New | Crop: Corn, silage | Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.44	62.20	367.88	3,580.55
Fresh water	10.99	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.44	62.20	367.88	3,580.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	228.70	67.26	289.23	1,139.73
Nutrient balance	75.74	-5.06	78.65	2,440.82
Applied to removed ratio	1.33	0.92	1.27	3.14

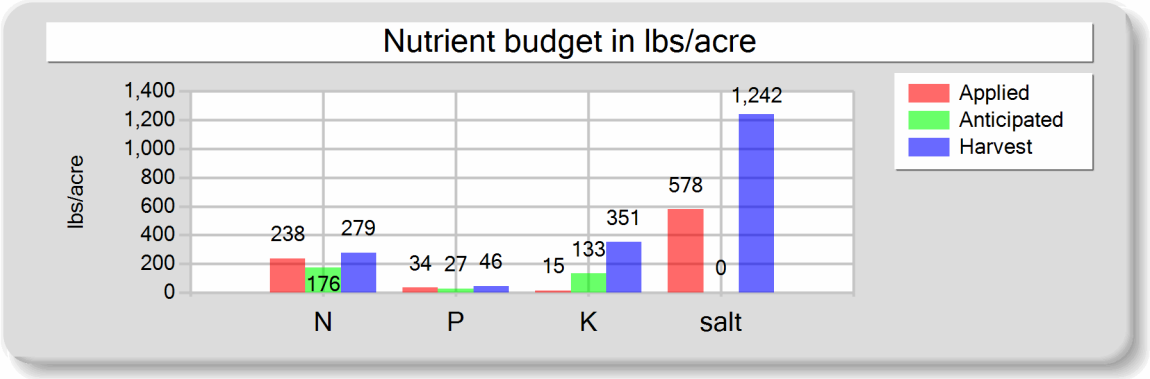
Fresh water applied	
16,830,000.00	*gallons*
619.79	*acre-inches*
26.95	*inches/acre*

Process wastewater applied	
3,870,000.00	*gallons*
142.52	*acre-inches*
6.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tack - 10/29/2022: Wheat, silage, soft dough

Field name: Tack | Crop: Wheat, silage, soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	87.07	10.34	14.72	0.00
Process wastewater	138.57	23.62	0.19	578.31
Fresh water	5.63	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.27	33.96	14.91	578.31
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	278.57	45.79	351.07	1,242.41
Nutrient balance	-40.30	-11.84	-336.16	-664.10
Applied to removed ratio	0.86	0.74	0.04	0.47

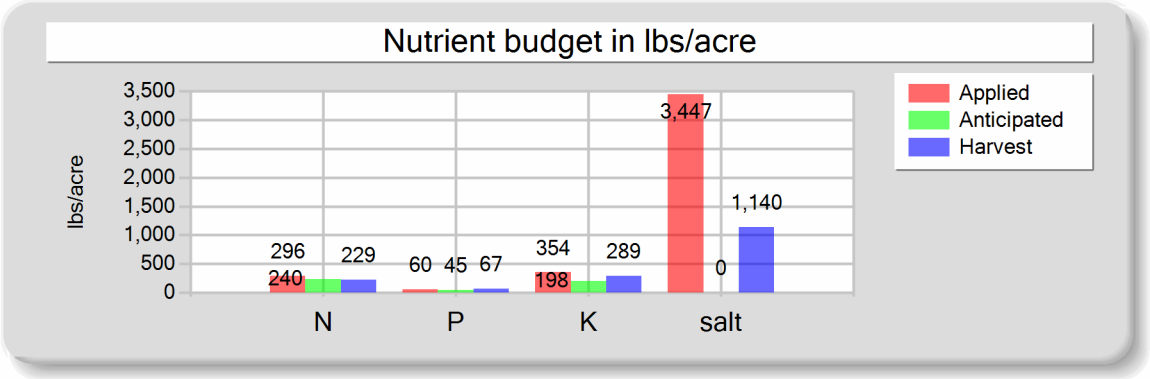
Fresh water applied	
1,500,000.00	*gallons*
55.24	*acre-inches*
13.81	*inches/acre*

Process wastewater applied	
180,000.00	*gallons*
6.63	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tack - 06/24/2023: Corn, silage

Field name: Tack | Crop: Corn, silage | Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.79	59.89	354.20	3,447.32
Fresh water	12.84	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.63	59.89	354.20	3,447.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	228.68	67.26	289.22	1,139.67
Nutrient balance	66.94	-7.37	64.98	2,307.65
Applied to removed ratio	1.29	0.89	1.22	3.02

Fresh water applied	
3,420,000.00	*gallons*
125.95	*acre-inches*
31.49	*inches/acre*

Process wastewater applied	
648,000.00	*gallons*
23.86	*acre-inches*
5.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Settling Pond Manure

Sample and source description: Settling Pond Manure

Sample date: 09/16/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	3,300.00	4,700.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 06/29/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	6,900.00	31,500.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 11/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	7,700.00	22,800.00	14,000.00	7,100.00	5,100.00	4,900.00	8,200.00		35.63
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	495.30	236.40	0.00	0.00	22.30	21.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,940.00	3,564
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	369.00	280.00	0.00	0.00	62.90	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,320.00	1,540
DL	10.00	2.00	0.50	0.50	0.20	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	0.10	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	204.00	179.00	0.00	0.00	44.30	262.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,840.00	2,550
DL	10.00	2.00	0.50	0.50	0.20	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	0.10	10

WWQ4

Sample and source description: WWQ4

Sample date: 11/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	236.00	226.00	0.00	0.00	54.60	249.00	107.00	91.20	107.00	35.60	0.00	27.80	7.10	4,600.00	3,050
DL	10.00	2.00	0.50	0.50	0.20	0.50	2.00	2.00	2.00	5.00	5.00	2.00	2.00	0.10	10

C. FRESH WATER ANALYSES

MID Lateral 5

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.10	37
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	10.00	10

River

River Orchard Well

Sample description: River Orchard Well

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	980.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

River Water

Sample description: River Water

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	110.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/27/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,000.00	8,000.00		9.23
DL	100.00	100.00	100.00		0.01

1 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,300.00	800.00	3,700.00		4.12
DL	100.00	100.00	100.00		0.01

4 - 10/30/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,200.00	11,200.00		6.11
DL	100.00	100.00	100.00		0.01

4 - 10/30/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,100.00	11,500.00		5.32
DL	100.00	100.00	100.00		0.01

4 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	800.00	5,000.00		6.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	700.00	4,400.00		6.14
DL	100.00	100.00	100.00		0.01

5 - 10/27/2022: Wheat, silage, soft dough

5 - 10/27/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,500.00	1,200.00	10,900.00		9.53
DL	100.00	100.00	100.00		0.01

5 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,800.00	800.00	4,000.00		4.82
DL	100.00	100.00	100.00		0.01

Big - 10/28/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,700.00	1,500.00	7,700.00		9.54
DL	100.00	100.00	100.00		0.01

Big - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	800.00	4,500.00		4.87
DL	100.00	100.00	100.00		0.01

Front - 10/29/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	1,200.00	5,900.00		7.14
DL	100.00	100.00	100.00		0.01

Front - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	900.00	4,000.00		4.63
DL	100.00	100.00	100.00		0.01

New - 10/29/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	900.00	5,200.00		8.42
DL	100.00	100.00	100.00		0.01

New - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	1,000.00	4,300.00		4.94
DL	100.00	100.00	100.00		0.01

Tack - 10/29/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	1,200.00	9,200.00		7.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

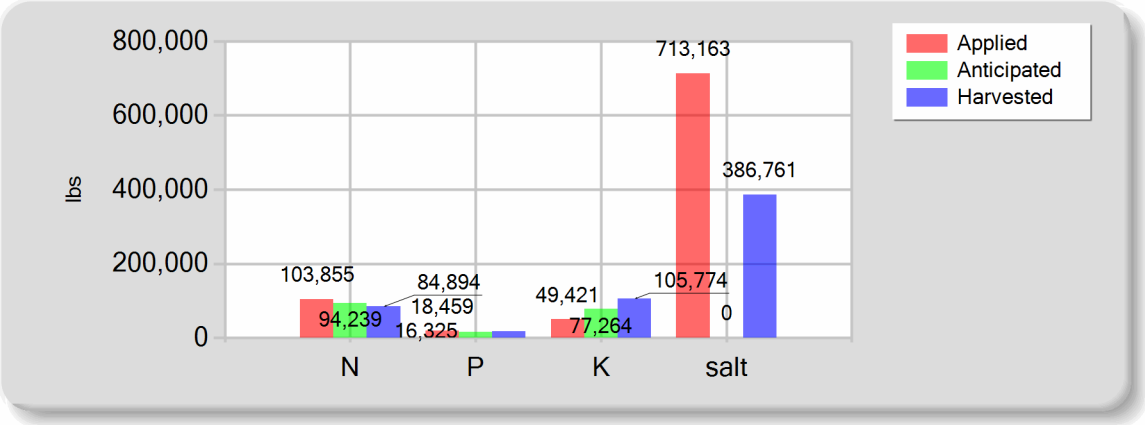
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

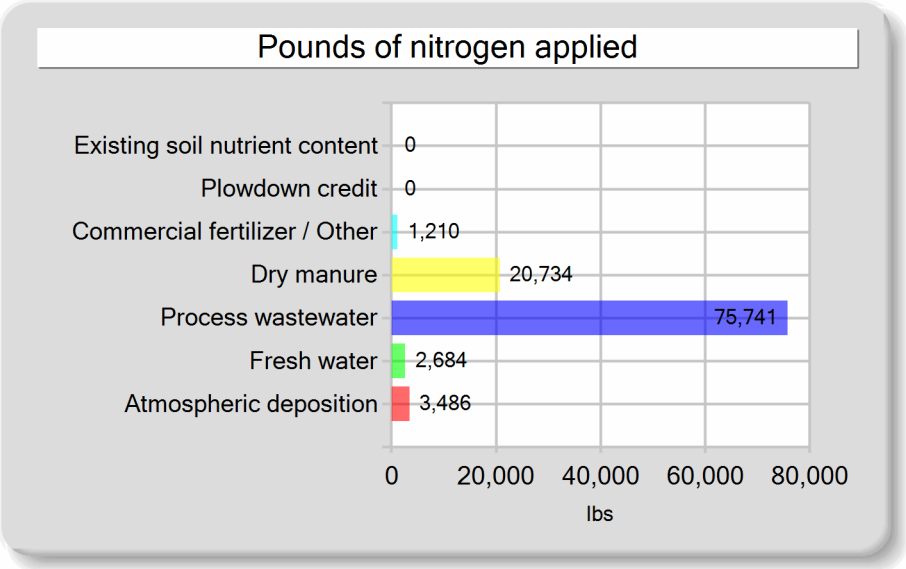
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

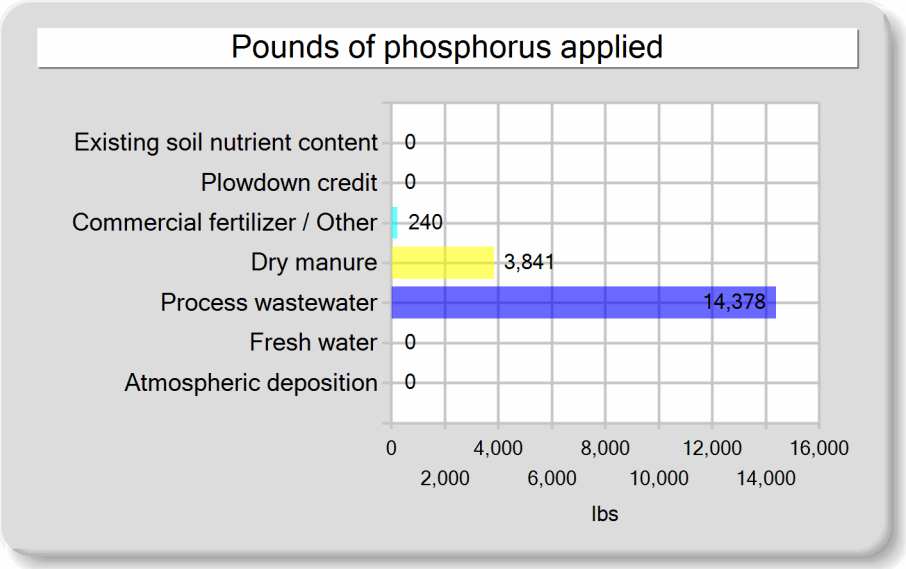
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,210.10	240.46	135.96	0.00
Dry manure	20,733.97	3,841.49	8,886.32	124,355.83
Process wastewater	75,741.05	14,377.55	40,398.97	577,628.98
Fresh water	2,683.90	0.00	0.00	11,178.06
Atmospheric deposition	3,486.00	0.00	0.00	0.00
Total nutrients applied	103,855.02	18,459.50	49,421.24	713,162.87
Anticipated crop nutrient removal	94,239.00	16,325.10	77,264.10	0.00
Actual crop nutrient removal	84,893.57	17,596.10	105,774.07	386,760.69
Nutrient balance	18,961.45	863.40	-56,352.83	326,402.18
Applied to removed ratio	1.22	1.05	0.47	1.84

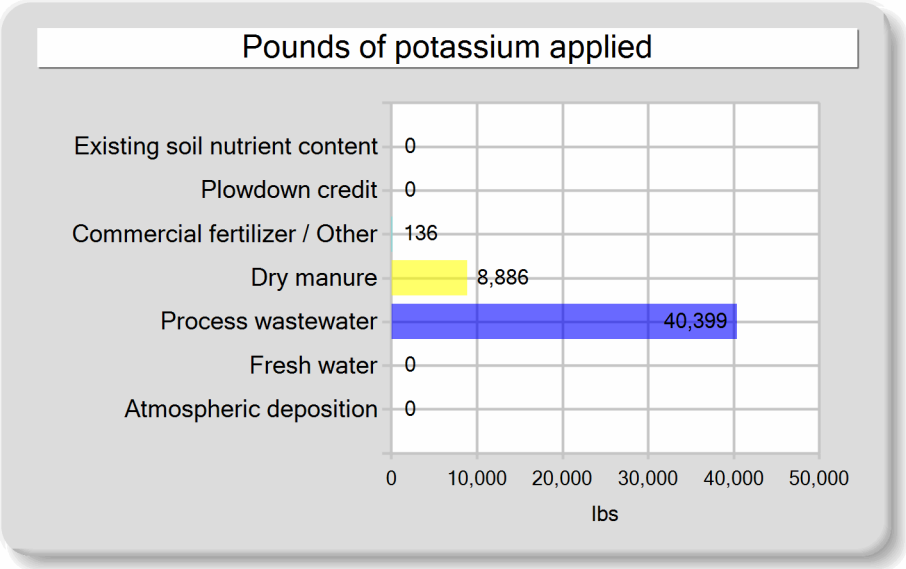
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

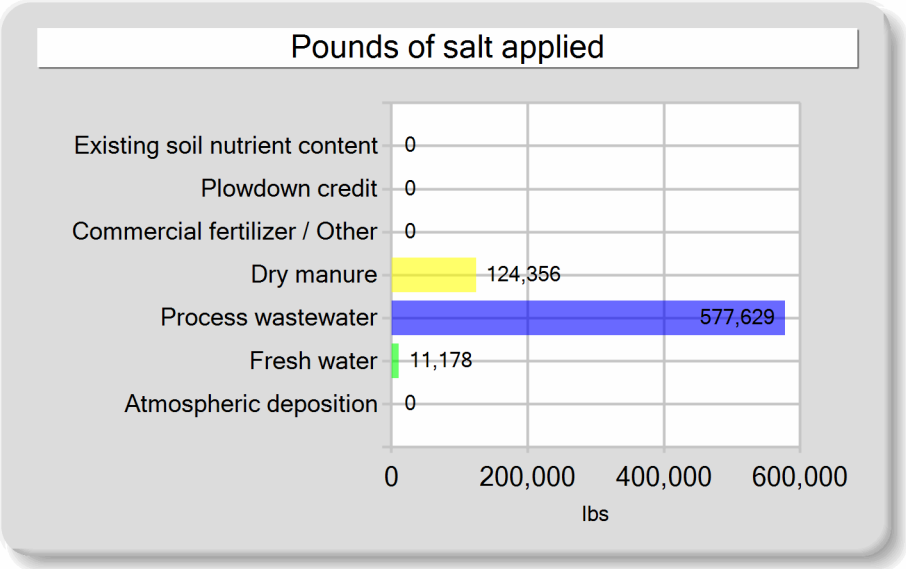


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 1 Corn (1.61), Front Field (1.49) and Big Field (1.51) all had high nitrogen field ratios as crops failed to meet the anticipated nitrogen removal of 240 pounds/acre. The subject fields removed 137.2, 161.5 and 165.1 pounds of nitrogen/acre. Yield for Field 1 Corn was adequate however a low nitrogen lab value from the plant tissue analysis resulted in only 4.6 pounds of nitrogen removed per harvested ton compared to the anticipated 8 pounds of nitrogen per harvested ton which resulted in the high nitrogen field ratio. With regard to Front Field and Big Field, yields were not as expected. The two fields produced only 17.6 and 23.6 tons/acre compared to the anticipated 30 tons/acre. This, along with a low nitrogen lab value on Big Field, resulted in the high nitrogen field ratio for the two fields. Nitrogen application to the three subject fields was not considered excessive but in-line with that recommended in the dairy's Nutrient Management Plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Couchman (Apr 3, 2024 16:04 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
David Brian Couchman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 3, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

10-26-23 Annual Well, River Water

A & L WESTERN LABORATORIES, INC.

1311 Woodland Ave. #1 • Modesto, California 95351 • Phone 209-529-4080

CHAIN OF CUSTODY

Client: Couchman Farms

Address: 3136 Iowa Ave

Modesto Zip 95358

Phone: (209) 315-9505 DAVE

CC to: vfortado agservices.com
couchmand59@gmail.com

Signature of person authorizing work under terms stated below*: David Couch

*Net 30 days. All accounts past due will be subject to interest charges of 1.5% per month.
*Hazardous materials are the property of the client. The client is responsible for proper disposal of hazardous wastes. Clients not picking up hazardous wastes may be assessed an appropriate fee.

PROJECT ID: Dairy Monitoring

SAMPLED BY: (signature): David Couch

Date	Time	sample type Grab	sample type Comp.	Sample Location	No. of Containers	ANALYSIS: Well WS								REMARKS
10-26-23	9:00	✓		Well Water #1 (New Milk Barn)	1		1							
		✓		Well Water #2 (Old Well)	1		1							
		✓		Well Water #3 DRY COW Pen	1		1							
				~~Lagoon Water~~	~~1~~		~~1~~							
		✓		River Water	1		1							
		✓		River Orchard Well	1		1							

Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
David Couch	10-26-23	9:00	[illegible]	10-26-23	11:00
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time
Relinquished by (signature):	Date	Time	Received by (signature):	Date	Time

Site Time: Start: Finish: Total Hours:

Driving Time: Start: Finish: Total Hours:

Dedicated Exclusively to Providing Quality Analytical Services

A & L Western Laboratories, Inc.

1311 Woodland Avenue, Modesto CA 95351 209-529-4080
10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

Ag Suitability Water Analysis

COUCHMAN FARMS
3136 IOWA AVENUE
MODESTO, CA 95358

Order Number: 70754
Lab Number: W70754-01
Submitted Date: 10/26/2023
Report Date: 11/1/2023
Submitted By: DAVID COUCHMAN

Grower: DAIRY MONITORING
Location: NEW MILK BARN
Description: WELL WATER #1

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Electrical Conductivity				Normal Values	Problem Values
Electrical Conductivity (EC)		1.50			
Results	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Nitrate - Nitrogen (NO_3-N)	32.9	2.35	88.8	<10	10>

RED = Value of Concern

contact@vtaglab.com

A & L Western Laboratories, Inc.

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Ag Suitability Water Analysis

COUCHMAN FARMS
3136 IOWA AVENUE
MODESTO, CA 95358

Order Number: 70754
Lab Number: W70754-02
Submitted Date: 10/26/2023
Report Date: 11/1/2023
Submitted By: DAVID COUCHMAN

Grower: DAIRY MONITORING
Location: OLD WELL
Description: WELL WATER # 2

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Electrical Conductivity				Normal Values	Problem Values
Electrical Conductivity (EC)		1.05			
Results	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Nitrate - Nitrogen (NO_3-N)	34.6	2.47	93.4	<10	10>

RED = Value of Concern

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Ag Suitability Water Analysis

COUCHMAN FARMS
3136 IOWA AVENUE
MODESTO, CA 95358

Order Number: 70754
Lab Number: W70754-03
Submitted Date: 10/26/2023
Report Date: 11/1/2023
Submitted By: DAVID COUCHMAN

Grower: DAIRY MONITORING
Location: DRY COW PEN
Description: WELL WATER #3

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Electrical Conductivity

				Normal Values	Problem Values
Electrical Conductivity (EC)		1.36			
Results	mg/L	meq/L	lbs/ac-ft	Normal Values mg/L	Problem Values mg/L
Nitrate - Nitrogen (NO_3-N)	21.3	1.52	57.5	<10	10>

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Ag Suitability Water Analysis

COUCHMAN FARMS
3136 IOWA AVENUE
MODESTO, CA 95358

Order Number:	70754
Lab Number:	W70754-04
Submitted Date:	10/26/2023
Report Date:	11/1/2023
Submitted By:	DAVID COUCHMAN

Grower: DAIRY MONITORING
Location: NEW MILK BARN
Description: RIVER WATER

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Electrical Conductivity				Normal Values	Problem Values
Electrical Conductivity (EC)		0.11			
Results	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Nitrate - Nitrogen (NO_3-N)	1.8	0.13	4.8	<10	10>

RED = Value of Concern

contact@vtaglab.com

A & L Western Laboratories, Inc.

1311 Woodland Avenue, Modesto CA 95351 209-529-4080
10220 SW Nimbus Avenue Bldg. K-9, Portland OR 97223 503-968-9225

Ag Suitability Water Analysis

COUCHMAN FARMS
3136 IOWA AVENUE
MODESTO, CA 95358

Order Number: 70754
Lab Number: W70754-05
Submitted Date: 10/26/2023
Report Date: 11/1/2023
Submitted By: DAVID COUCHMAN

Grower: DAIRY MONITORING
Location: NEW MILK BARN
Description: RIVER ORCHARD WELL

mg/L = milligrams/liter = part per million = ppm
meq/L = milliequivalents/liter
dS/m = deciSiemen/meter = mmhos/cm
lbs/ac-ft = pounds/acre-foot

Electrical Conductivity				Normal Values	Problem Values
Electrical Conductivity (EC)		0.98			
Results	**mg/L**	**meq/L**	**lbs/ac-ft**	**Normal Values mg/L**	**Problem Values mg/L**
Nitrate - Nitrogen (NO_3-N)	33.3	2.38	90.0	<10	10>

RED = Value of Concern

contact@vtaglab.com

2023 Annual Report Couchman Farms

Final Audit Report 2024-04-03

Created:	2024-03-26
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAATmGdILHuD9ZvwZ4puQ3yh7lymwzFaIhN

"2023 Annual Report Couchman Farms" History

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Signer couchmand59@gmail.com entered name at signing as David Couchman
2024-04-03 - 11:04:32 PM GMT

Document e-signed by David Couchman (couchmand59@gmail.com)
Signature Date: 2024-04-03 - 11:04:34 PM GMT - Time Source: server

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